DSA 210 Final Report

Motivation

The goal of this project was to apply data science to my personal life, specifically my workout life. Beginning with the spring semester, I started waking up and going to gym earlier so that I can save time for other stuff during the day. Doing so, I wondered whether my shift to a productive day translates to productivity in the gym.

Data Source

I had been tracking my workouts for a year using the workout tracking application called Hevy, and the application provides the ability to export your data as a .csv file.

Data Analysis

All analyses were conducted in Python.

1. **Cleaning**

Datetimes were formatted, spring and fall semester were separated, and missing weight values (pull ups, dips) were replaced with my weight (81kg).

2. **Engineering**

I made my target metric ***volume_per_min*** (training efficiency). I also made made other metrics to come to a conclusion for my question such as ***Total_Days_Lifting*** (tracking experience) and ***Hours_Since_Last_Workout*** (tracking recovery time)

3. **Modeling**

I used Linear Regression and an Independent T-test to evaluate rate of growth (***volume_per_min*** compared to days since start of the semester) and Random Forest Regressor to predict future workouts.

Findings

Hypothesis:

I had initially hypothesized that my spring semester, where I consistently went to gym in the mornings, would yield a higher rate of growth compared to my fall semester, where I went to gym inconsistently and at random times. However, linear regression proved that the Fall Semester actually had a steeper rate of growth (+0.28 kg/min per day vs +0.11 kg/min per day).

Machine Learning:

The Random Forest Regressor predicted my workouts with a Mean Absolute Error (MAE) of just 16.44 kg/min. The Feature Importance graph showed that ***Total_Days_Lifting*** was the most dominant feature in predicting future workouts, followed by ***Hours_Since_Last_Workout***, which prove that total experience, and recovery time matter significantly more than the time of day.

Limitations and Future Work

While the Fall Semester had a better rate of growth, there are 2 confounding variables to be taken into account.

First of all, the Fall semester followed a 4 month break from the gym, meaning the workouts in the Fall semester were regaining lost muscle, which is proven to be easier to accomplish than gaining new muscle, which is what I was doing in the Spring semester.

Secondly, during the Fall semester, I was taking Creatine daily which I stopped when beginning the Spring semester.

Because these variables are exclusive to the Fall semester, the model can’t actually determine whether the steeper slope in Fall was caused by the schedule, or the supplementation and muscle memory.

Future extensions of this project would have to include A/B testing, where diet and supplementation are kept the same to isolate the Time of Day variable perfectly.